COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE010/Data Structures and Algorithms | **SECTION:** | CPE21S3 |
| **DATE PERFORMED:** | Jul 16, 2026 | **DATE SUBMITTED:** | |
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**ACTIVITY NO. 1** **Arrays, Pointers and Dynamic Memory Allocation**

## 1. PROCEDURE OUTPUT

Screenshot

CPE010_HOA3p1_Panti.cpp > main()

```cpp
#include <iostream>
#include <string.h>

class Student{
  private:
  std::string studentName;
  int studentAge;

 public:
Student(std::string newName ="John Doe", int newAge=18){
studentName = std::move(newName);
studentAge = newAge;
std::cout << "Constructor Called." << std::endl;
};
  ~Student(){
std::cout << "Destructor Called." << std::endl;
}
Student(const Student &copyStudent){
std::cout << "Copy Constructor Called" << std::endl;
studentName = copyStudent.studentName;
studentAge = copyStudent.studentAge;
}
void printDetails(){
std::cout << this->studentName << " " << this->studentAge << std::endl;
}
};

int main() {
Student student1("Roman", 28);
Student student2(student1);
Student student3;
student3 = student2;

return 0;
}
```

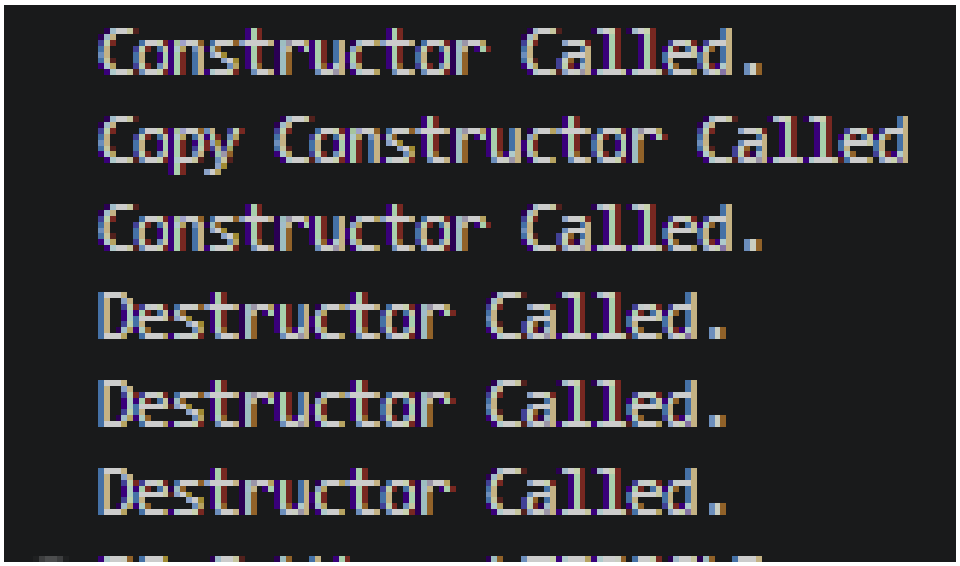
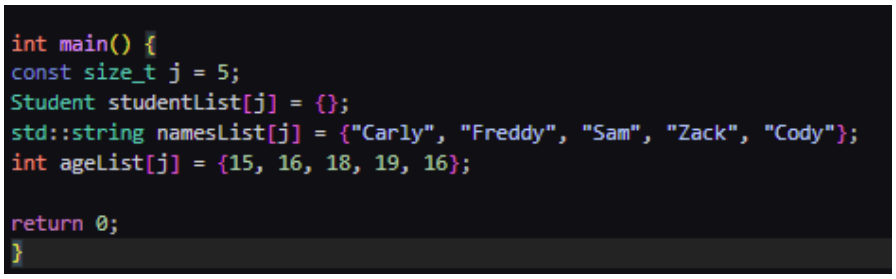
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| Observation | In student1, the constructor initializes the values within the main code and processes them accordingly. In student2, it copies the parameters in student1 and allocates them to itself. While student3 initializes the process in student2, creating a chain of pointers to the values of student1. |

Table 1: Initial Driver Program

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| Screenshot |  |

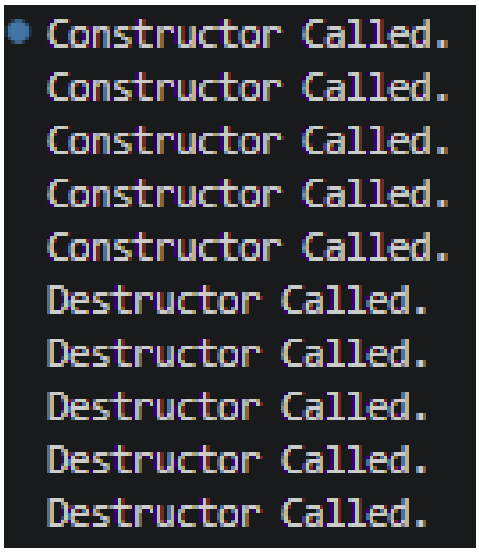
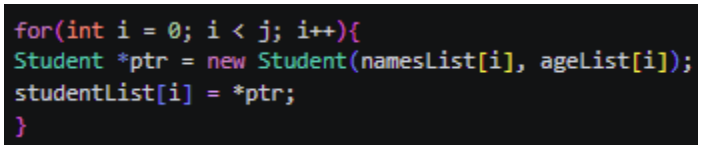
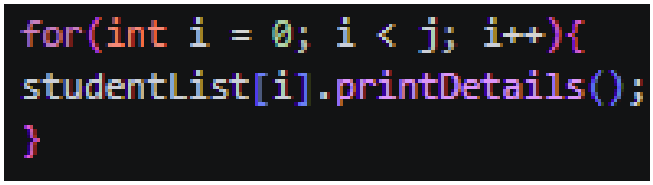
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| Observation | It creates an array labeled studentList with a constant size of 5. Then it initializes namesList and ageList arrays with given values. |

Table 2: Modified Driver Program with Student Lists

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|---|---|
| Loop A |  |
| Observation | In the loop, it stored the Student's nameList and ageList arrays in a pointer which it copies to the studentList array, repeating the constructor 5 times. |
| Loop B |  |
| Observation | This loop initializes the contents of studentList, iterating each value until the 5th. |

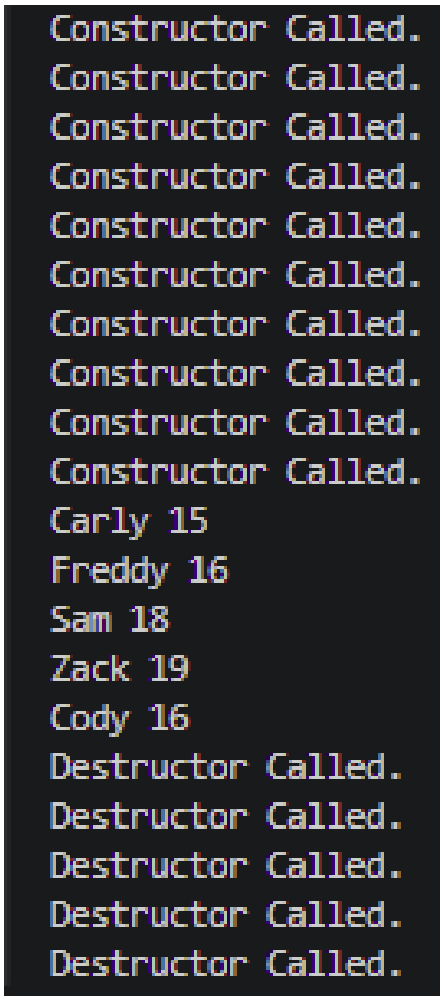
| | |
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| Output |  |
| Observation | The constructors from table 2 initialize the 2 arrays with 5 values each, initializing 5 times. Then the first loop constructs the studentList array which stores the values from the 2 arrays, which then initializes the second loop to start printing the values of studentList. And lastly, using destructors in the function, wipes the allocated values after the program finishes. |

Table 3: Final Driver Program

Modifications

```cpp
#include <iostream>
#include <string.h>

class Student{
  private:
  std::string studentName;
  int studentAge;

 public:
Student(std::string newName ="Joma", int newAge=67){
studentName = std::move(newName);
studentAge = newAge;
std::cout << "Constructor Called." << std::endl;

};
  ~Student(){
std::cout << "Destructor Called." << std::endl;
}
Student(const Student &copyStudent){
std::cout << "Copy Constructor Called" << std::endl;
studentName = copyStudent.studentName;
studentAge = copyStudent.studentAge;
}
void printDetails(){
std::cout << this->studentName << " " << this->studentAge << std::endl;
}
};

int main() {
const size_t j = 5;
Student studentList[j] = {};
std::string namesList[j] = {"Elaiza", "Fenzie", "Rodnee", "Nina", "Carlo"};
int ageList[j] = {21, 19, 18, 19, 20};

for(int i = 0; i < j; i++){
Student *ptr = new Student(namesList[i], ageList[i]);
studentList[i] = *ptr;
delete[] ptr;
ptr=nullptr;
}
for(int i = 0; i < j; i++){
studentList[i].printDetails();
}

return 0;
}
```

```
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Destructor Called.
```

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| Observation | In adding the destructor lines, the code runs smoothly through the system, allowing memory to be reallocated. |

Table 4: Modifications/Corrections Necessary

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **AI Tutor(Python Tutor), Grammarly**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **Explanation, idea generation, grammar correction**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **Explain code, give ideas, and correct grammar**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

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